Whisperways

Noise-aware eVTOL flight corridors: route flight by who hears it.

HTCJ Aviation Futures Innovation Challenge
Track: Best Advanced Air Mobility Innovation
July 2026 · Dan Mercede, built with Claude
live: **whisperways.vercel.app**
code: github.com/OrionArchitekton/whisperways

The problem: AAM is gated by noise, and nobody routes for it

Advanced air mobility will not be decided by range or battery chemistry. It will be decided by community acceptance, and the gating complaint is noise: it drives vertiport approvals, certification limits, and neighborhood opposition. Yet corridor planning today optimizes distance and airspace. The people underneath the flight path are not in the objective function. Whisperways puts them there.

What it does

Pick two Los Angeles vertiports. The engine returns the **direct route** (distance-optimal) and the **quiet route** (population-noise-optimal), each with distance, flight time, the number of people who can hear the aircraft above ambient, and a population-weighted exposure index. Claude then drafts the **Community Impact Brief**: a neighborhood-by-neighborhood account of what each place hears under each corridor, with the tradeoff stated honestly, grounded only in engine-computed numbers.

221,225	50,750	-88.8%
people hear the DIRECT route: LAX to Hollywood Burbank, night ops, 8.4 min	people hear the QUIET corridor the engine found: 12.7 min	population-weighted exposure index (77% fewer people audible), for +4.3 minutes

Architecture

POPULATION US Census Bureau 2020 tract centers of population (LA County, public domain), rasterized to a 250 m grid: 185x156 cells, 1,408 tracts, 5.32M people. Build script and provenance metadata committed with the artifact.

ACOUSTICS First-order model: published eVTOL overflight level (Joby S4, ~45.2 dBA at 500 m, NASA/Joby acoustic flight campaign, results published 2022), geometric spreading L(d) = L0 - 20 log10(d/d0), day/night ambient baselines. Labeled in the UI as a planning heuristic, not certification acoustics.

ROUTING Population is convolved with the audible-footprint kernel into a person-decibel cost raster; Dijkstra then trades meters of detour against person-dB via a single bias weight. Exposure accounting is footprint-based: each ground cell hears only its loudest pass.

AI BRIEF One server route calls claude-opus-4-8 with a JSON-schema-constrained output. The prompt receives only engine output (places, levels, tradeoffs) and forbids inventing numbers; the response is shape-validated, length-capped, and scrubbed defensively.

DELIVERY Next.js App Router + MapLibre GL on Vercel. API key server-side only. The AI endpoint is rate-limited through Upstash Redis, and the limit is burst-verified (concurrent 10-burst returns 5x200 + 5x429).

The differentiator

AAM hackathon entries cluster on booking apps, delivery-route optimizers, and delay dashboards. Whisperways is **noise-equity tooling**: it treats community acceptance as the engineering constraint it actually is, computes who bears the noise burden of every corridor decision, and produces the document an operator would hand a neighborhood council. FAA integration planning ties community noise to the public acceptance it says will decide the industry, and NASA's UAM noise research names vertiport-era noise planning a major open gap. That gap is this product.

Engineering rigor

- **28 tests** (vitest): attenuation physics, kernel and raster properties, route optimality on synthetic grids, and real-city acceptance (the quiet route must beat direct exposure at bounded time cost on live census data).
- **Pure engine**: acoustics + routing are dependency-free TypeScript modules; the UI and API are thin consumers.
- **Grounded AI by construction**: schema-constrained output, engine-only inputs, defensive parse: engine-only inputs, a fixed output schema, and a prompt that forbids invented numbers.
- **Abuse guard proven, not assumed**: the rate limit was burst-tested live until it returned 429s.
- **Spec-first**: behavior contract in `specs/whisperways-spec.md`; demo determinism via a disclosed frozen capture of a real run.

`TypeScript` `Next.js 16` `MapLibre GL` `claude-opus-4-8` `Vercel` `Upstash` `Census 2020` `vitest x28`

Challenges

- **Cruise eVTOLs are quiet.** Published overflight levels sit near daytime urban ambient, so the demo scenario is night residential operations, exactly when real noise complaints gate approvals.
- **Surrogate vs metric.** Dijkstra minimizes a path-sum cost while humans experience footprint exposure; property tests pin the two together across vertiport pairs.
- **Honest AI on thin data.** Early briefs had no neighborhoods to talk about because the quiet route avoided them all; the fix was reporting what each place gains between corridors, not just what it suffers.

HONESTY NOTES

First-order acoustics (no atmospheric absorption, terrain, or directivity): numbers are planning-grade comparisons between routes, not certifications. Vertiport set is curated and illustrative. Demo mode serves a frozen result for deterministic capture; it is a genuine output of the same engine and live Claude path, disclosed in the README. Built end-to-end with Claude (Claude Fable 5 in Claude Code); the git Co-Authored-By trailers record it.

Judge it in 60 seconds

1. Open **whisperways.vercel.app** and press **Plan corridors**: the amber direct route and teal quiet corridor land on the map with live metrics.
2. Read the hero card: exposure reduction versus added minutes, computed from census population.
3. Press **Generate community impact brief**: Claude writes the neighborhood-by-neighborhood brief live from the engine's numbers.